

EDWARD HONG



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Education

University of Waterloo

September 2022 - Present

Honours Nanotechnology Engineering | Waterloo, ON

- Wrote technical lab reports on affects of nanofillers on nanocomposites' mechanical and thermal properties
 - Conducted impact and tensile tests, Differential Scanning Calorimetry, Thermogravimetric Analysis, and Atomic Force Microscopy, isolate target products from solutions
 - Synthesized nanosilver particles to determine its effectiveness against elimination of micro-organisms
- Knowledge of circuits and electromagnetism concepts
 - Conducted dry labs by utilizing COMSOL simulation, oscilloscope, soldering, and circuit building
- Characterized materials through variety of instrumental methods
 - Utilized Scanning Electron Microscopy to visualize topography and determine composition of semiconductors
 - Synthesized MEMs on silicon wafers through lithography to perform Ellipsometry

Experience

Research Assistant

May 2026 - September 2026

Irradiant Technologies Inc. | Waltham, MA, USA

- Assisted in synthesis and characterization of samples at Harvard and MIT
- Took lead and implemented automation robot for overnight lab operations to reduce turn around time of contract deliverables
- Researched varying novel methods of 3D printing structures using Two Photon Lithography (TPL)
- Utilized the X-Tool for creating patterns and samples with laser ablation
- Modified and printed CAD models to be used for automated pipetting

Research Assistant

Jan 2026 - April 2026

York University | Toronto, ON

- Assisted in conducting Carbon Capture Utilization Storage (CCUS) experiments using customized Electrochemically Mediated Amine Regeneration (EMAR) system.
- Assembled the apparatus and tested Nanoparticle Tracking Analysis (NTA) using Dynamic Light Scattering (DLS).
- Conducted literature research for EMAR and NTA.
- Supported keeping stock of lab supplies for daily operation.

Material Science Specialist

September 2024 - April 2025

H2nanO | Kitchener, ON

- Lead a project on synthesizing hydrophilic coatings on photocatalysts
- Compiled and analyzed data, reviews and project plans for weekly meetings
- Designed and printed 3D CAD models for instrumentation (contact angle goniometer)
- Conducted multiple material characterizations through numerous methods (UV - Vis, SEM, FTIR, COD) for semiconductors and nanoparticles
- Processed weather data through collaborative Python projects

Lab Assistant

January 2024 - April 2024

SMART - Lab, University of Waterloo | Waterloo, ON

- Co - authored in published research paper
- Participated in multiple projects to assist in the creation of hydrogels for the purpose of soft body robotics
- Synthesized and characterized hydrogels with nanocomposites (Crystal Nanocellulose, Iron Oxide nanoparticles)
- Designed and printed 3D CAD models for extrusion 3D printing, as well as DLP 3D printing
- Shape-change programming of hydrogels through a gradient in chemistry for soft robotic applications
- Debriefed projects during conference meetings to ensure intercommunication skill
- Took stock of chemicals and created product comparison charts between different providers

Skills

- Proficient in Python, MATLAB, and C++
- Fluent in English and Korean

Lab Equipment

- COMSOL Simulation
- DC Sputtering System, MJB Mask Aligner, Raman